

Glimpse Guide

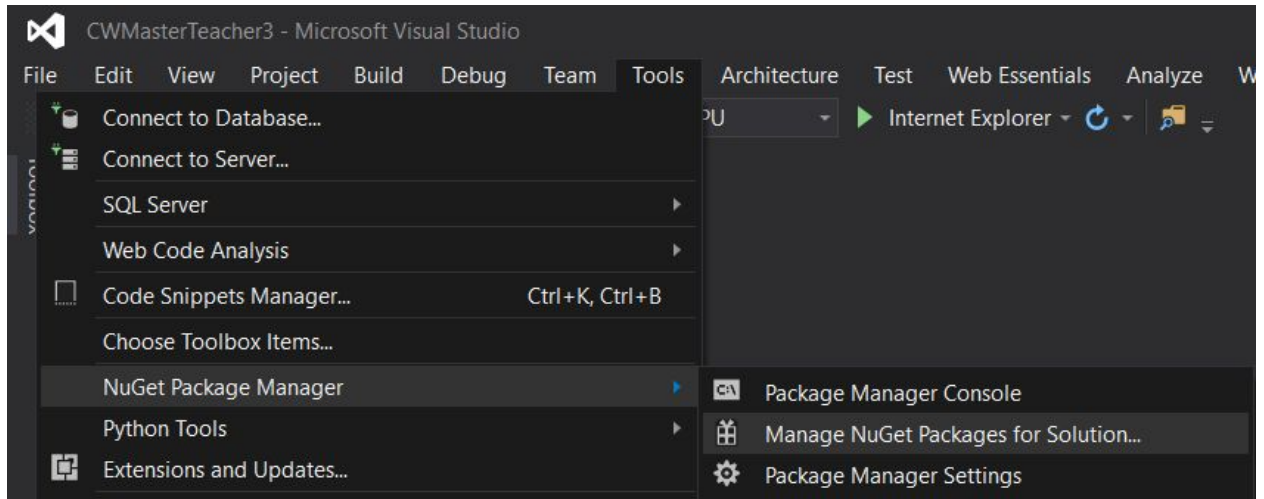
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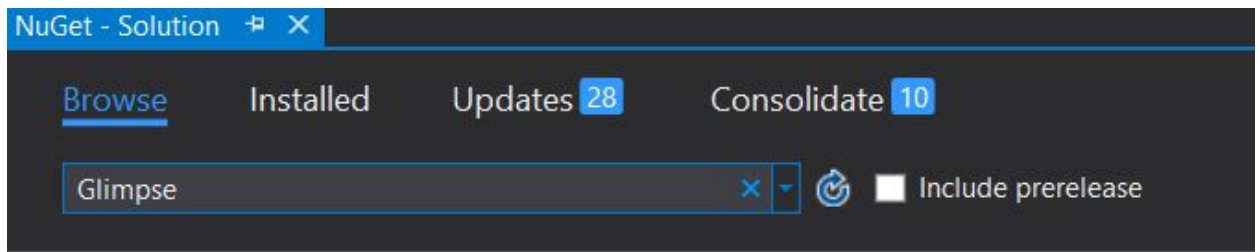
Installing Glimpse

Installing Glimpse using the NuGet Manage Packages Interface

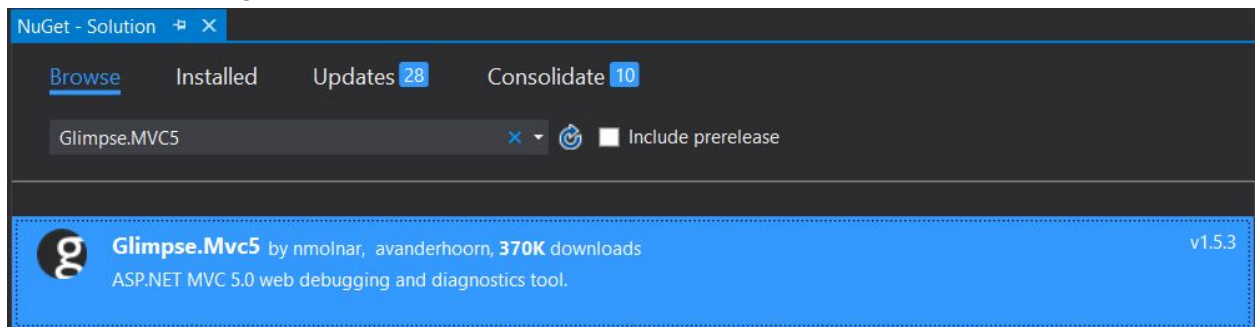
1. Open Microsoft Visual Studio 2013/2015 and open the CW Master Teacher project
2. Navigate to the toolbar at the top of the application and to the tab Tools→ NuGet Package Manager → Manage Packages for Solution



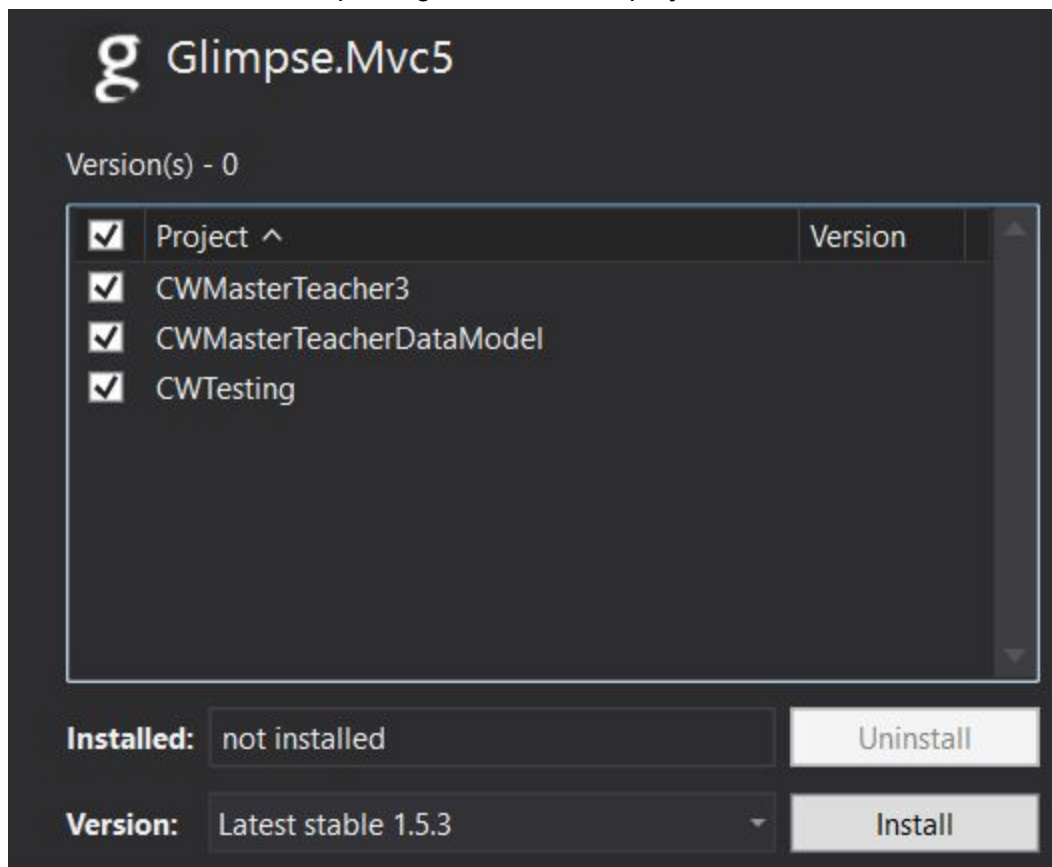
3. Left-Click on the Browse option and enter Glimpse in the search field



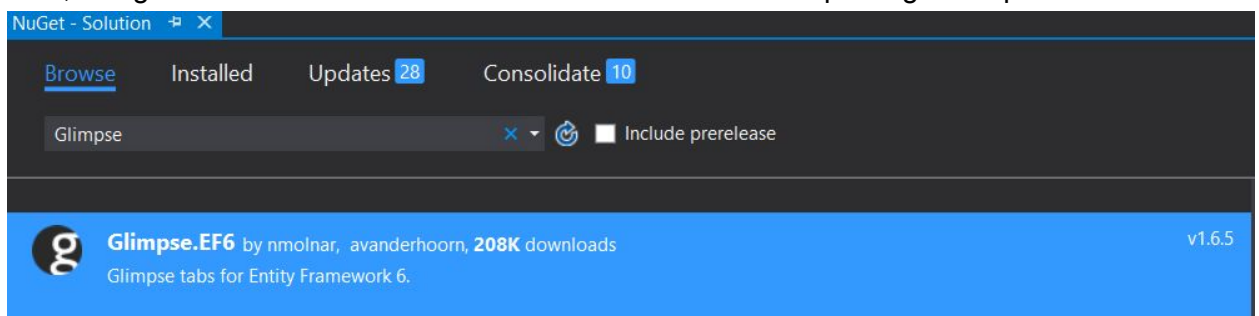
4. Choose the package Glimpse.Mvc5



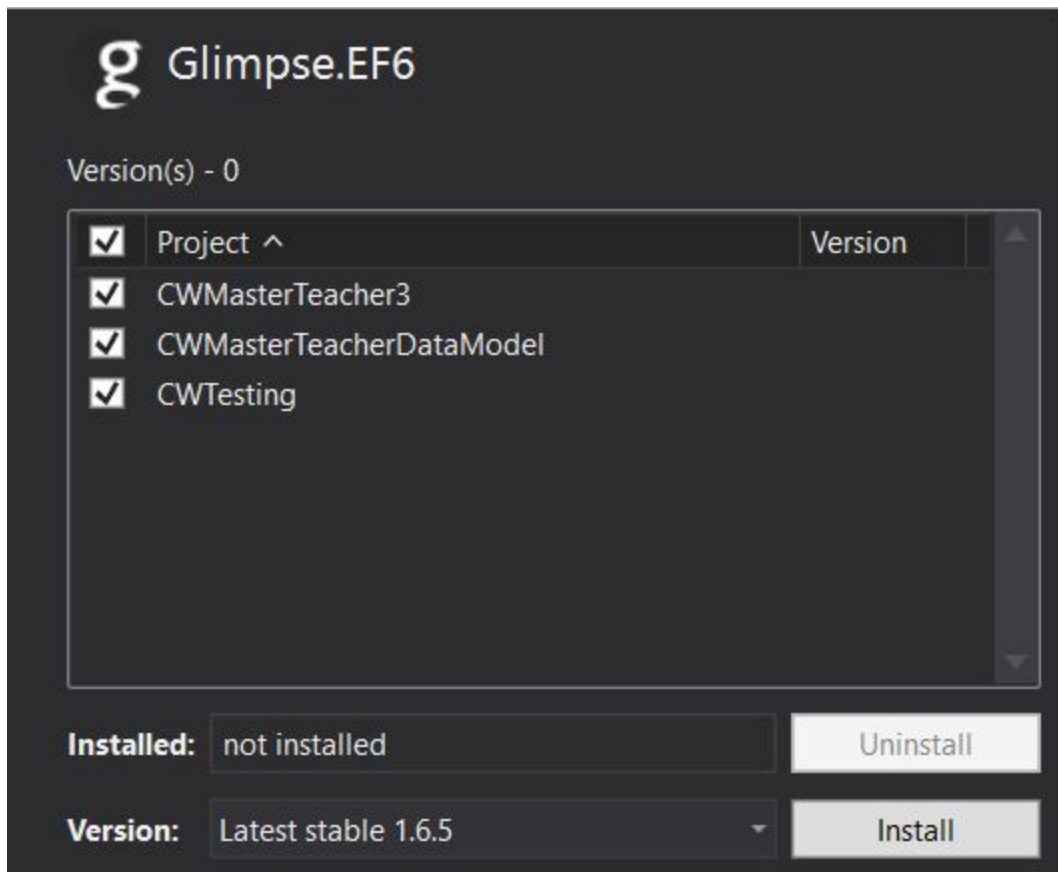
5. On the right hand side select the Project checkbox which will autoselect the remaining checkboxes to install the package for the entire project



6. In the Version dropdown menu select the Latest stable option. At the time this guide was written that was: Latest stable 1.5.3
7. Left-Click install. Installation may take a few minutes. When installation is finished a Readme-Glimpse tab will be opened displaying a message thanking you for installing Glimpse
8. Next, navigate back to the NuGet -Solution tab and choose the package Glimpse.EF6



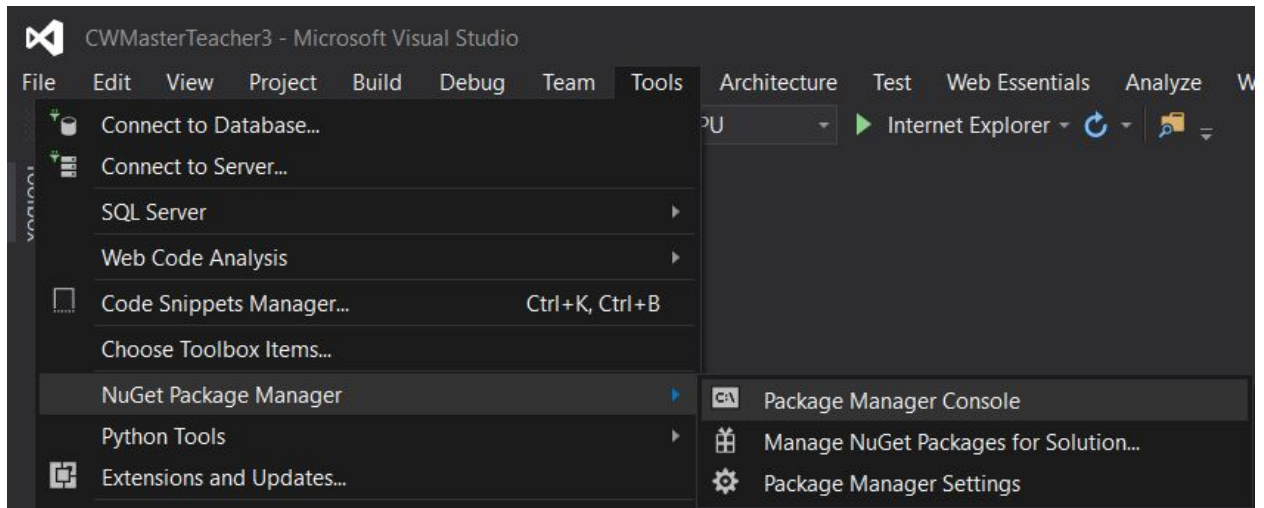
9. On the right hand side select the Project checkbox which will autoselect the remaining checkboxes to install the package for the entire project.



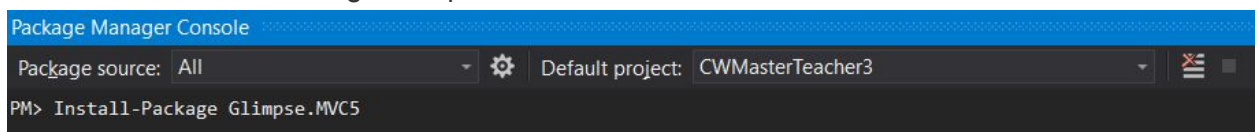
10. In the Version dropdown menu select the Latest stable option. At the time this guide was written that was: Latest stable 1.6.5.
11. Left-Click install. Installation may take a few minutes. When installation is finished a Readme-Glimpse tab will be opened displaying a message thanking you for installing Glimpse. Congratulations, you now have Glimpse installed and ready to run!

Installing Glimpse using the NuGet Package Manager Console

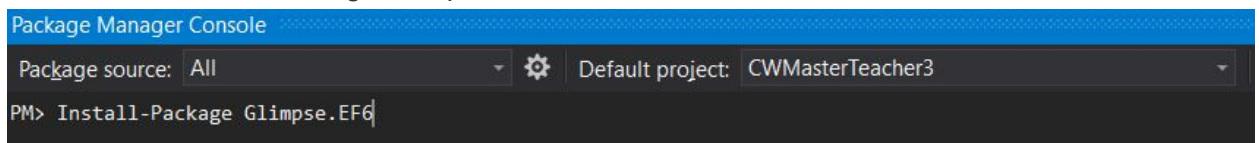
1. Open Microsoft Visual Studio 2013/2015 and open the CW Master Teacher project.
2. Navigate to the toolbar at the top of the application and to the tab Tools→ NuGet Package Manager→ Package Manager Console



3. At the prompt "PM>" in the package manager console enter:
"PM> Install-Package Glimpse.MVC5"

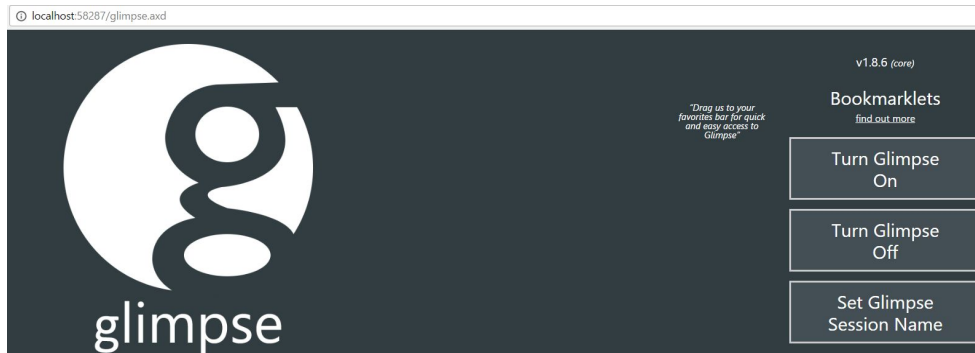


4. Once the MVC5 package has finished installing, install the Entity Framework package by entering:
"PM> Install-Package Glimpse.EF6"



Activating Glimpse

1. Run the CW Master Teacher project from inside Microsoft Visual Studio using the browser of your choice.
2. Inside the browser navigate to /Glimpse.axd, for example:
(<http://localhost:1234/glimpse.axd>) NOTE: The actual local host number is execution dependant, simply use the same local host that is displayed in the original URL.



3. Click on the button "Turn Glimpse On"
4. Return to the CW Master Teacher homepage and begin performing the actions which are the focus of the performance diagnostics.

Automatic Activation Configuration

1. Allow Glimpse to run automatically each time the instance of CW Master Teacher on which Glimpse is installed runs.
2. Open Solution Explorer in Visual Studio and navigate to the project's Web.config file in CW Master Teacher
3. Inside the Web.config file, navigate to the bottom of the file and locate the Glimpse section.

Add the following lines into the Glimpse section:

```
<runtimePolicies>
  <ignoredTypes>
    <add type="Glimpse.Core.Policy.ControlCookiePolicy, Glimpse.Core"/>
  </ignoredTypes>
</runtimePolicies>
```

Example

```
219 <glimpse defaultRuntimePolicy="On" endpointBaseUri="~/Glimpse.axd">
220   <!--
221     For more information on how to configure Glimpse, please visit http://getglimpse.com/Help/Configuration
222     or access {your site}/Glimpse.axd for even more details and a Configuration Tool to support you.
223   -->
224   <runtimePolicies>
225     <ignoredTypes>
226       <add type="Glimpse.Core.Policy.ControlCookiePolicy, Glimpse.Core"/>
227     </ignoredTypes>
228   </runtimePolicies>
229 </glimpse></configuration>
```

Configuration for Remote Server Deployment

1. Open Solution Explorer in Visual Studio and navigate to the project's Web.config file in CW Master Teacher
2. Inside the Web.config file, navigate to the bottom of the file and locate the Glimpse section.

Add the following lines into the Glimpse section:

```
<runtimePolicies>
  <ignoredTypes>
    <add type="Glimpse.AspNet.Policy.LocalPolicy, Glimpse.AspNet"/>
  </ignoredTypes>
</runtimePolicies>
```

Example

```
219 <glimpse defaultRuntimePolicy="On" endpointBaseUri="~/Glimpse.axd">
220   <!--
221     For more information on how to configure Glimpse, please visit http://getglimpse.com/Help/Configuration
222     or access {your site}/Glimpse.axd for even more details and a Configuration Tool to support you.
223   -->
224   <runtimePolicies>
225     <ignoredTypes>
226       <add type="Glimpse.AspNet.Policy.LocalPolicy, Glimpse.AspNet"/>
227     </ignoredTypes>
228   </runtimePolicies>
229 </glimpse></configuration>
```

HUD Diagnostics

The HUD feature within Glimpse is used to gather the breakdown of how much time is spent in each stage of processing a request as well as the relevant statistics concerning database access and time spent within a controller or action. The HUD allows you as a developer to quickly and easily monitor the performance of each action as it is performed.

1. Run the CW Master Teacher project from within Microsoft Visual Studio.
2. Activate Glimpse if necessary.
(If the Activation Required Policy has been disabled, Glimpse will activate automatically)
3. Begin interacting with the project's webpage. This will cause Glimpse to begin collecting data and displaying it in the HUD located at the bottom of the page.

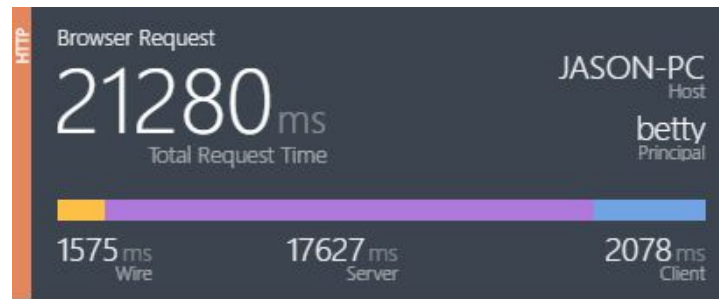


HTTP Section

The HTTP section of the HUD provides a breakdown of how much time was spent responding to the action taken by the actor. The following information is displayed.

- **wire:** time spent being transferred over the network including DNS lookup and time spent sending the request and receiving a response
- **server:** total time spent on the server including time being serviced by the controller and time spent retrieving records from the database

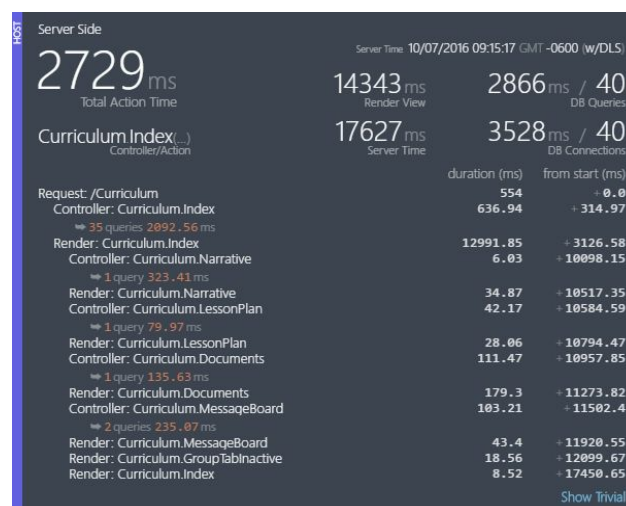
- **client:** time spent being interpreted and rendered by the client once the client kicks in to DOM ready.



Host Section

The Host section of the HUD provides a more detailed view of the actions occurring on the server. The following information is displayed.

- In the top left hand corner is the total time spent by the server while servicing this request.
- Directly below the total time is the controller and action which was called to service the request.
- Along the left hand side is the call chain for servicing the request along with the number of database queries if any occurred and the time they took to be resolved.
- In the top and center of the host display is the time it took the root view to render and the time it took the root action to execute.
- On the top right side is the number of database queries and connections along with the time it took them to execute.
- The "duration" column that corresponds to the call chain refers to the duration of each step in the chain.
- The "from start" column refers to when the event occurred in relation to the start of the request.



Main Panel Diagnostics

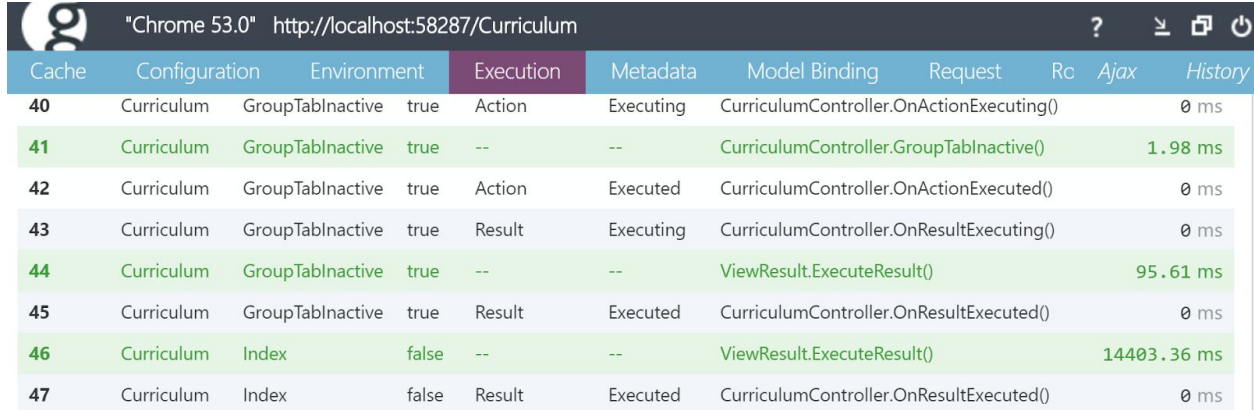
The main panel within Glimpse is used to gather detailed information about performance of the events that take place when servicing a request. These events include any executed controller and its action(s) as well as a deeper look at the structure, number, and execution time of all database queries. In order to pull up the main panel for Glimpse click on the "G" on the right side of the HUD at the bottom of the page.

Execution Tab

The execution tab provides a detailed time footprint of the actions which take place while servicing a request. The layout of columns is, from left to right:

(List taken directly from Glimpse documentation)

- **Ordinal:** Indicates where in the execution pipeline this method occurred
- **Controller & Action:** Controller & Action that the execution statement belongs too
- **IsChild:** Whether this method was executed in the context of a child request
- **Category:** The Category/type of the filter. Valid values include: Authorization, Action, Response, Exception
- **Type.Method:** Type that the executed method belongs to and the name of the method executed
- **Time Elapsed:** The amount of time it took to execute the method



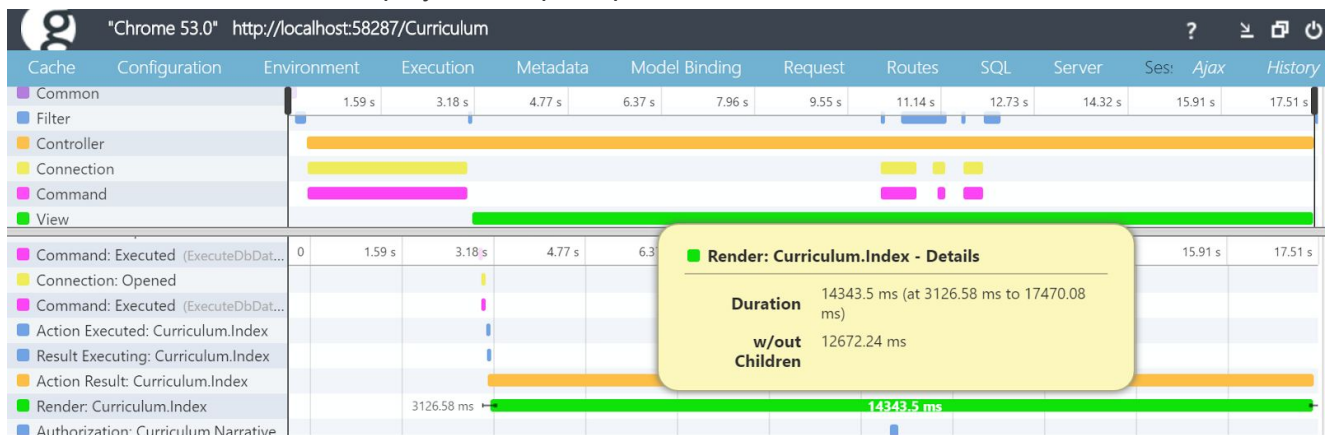
Cache	Configuration	Environment	Execution	Metadata	Model Binding	Request	Rc	Ajax	History
40	Curriculum	GroupTabInactive	true	Action	Executing	CurriculumController.OnActionExecuting()			0 ms
41	Curriculum	GroupTabInactive	true	--	--	CurriculumController.GroupTabInactive()			1.98 ms
42	Curriculum	GroupTabInactive	true	Action	Executed	CurriculumController.OnActionExecuted()			0 ms
43	Curriculum	GroupTabInactive	true	Result	Executing	CurriculumController.OnResultExecuting()			0 ms
44	Curriculum	GroupTabInactive	true	--	--	ViewResult.ExecuteResult()			95.61 ms
45	Curriculum	GroupTabInactive	true	Result	Executed	CurriculumController.OnResultExecuted()			0 ms
46	Curriculum	Index	false	--	--	ViewResult.ExecuteResult()			14403.36 ms
47	Curriculum	Index	false	Result	Executed	CurriculumController.OnResultExecuted()			0 ms

Timeline Tab

The Timeline tab provides a visual view of each action's type as well as when it occurred and how long it took. It also provides a synopsis of how much time was spent on each type of action. By consolidating the information from the other tabs it provides a comprehensive overview of the performance data.

- The time bar in the summary section displays both the duration of the event and the duration of the event if the duration of the children are not included. It also displays the start and end times of the event.

- There is also an alternate view of the timeline that turns the display into a list similar to the execution tab. This view is helpful to quickly scan for large duration times but sacrifices the visual display that helps to place the events in a relative context.



SQL Tab

The SQL tab provides a detailed look at the database queries performed to service the request. It is important to note that this view allows the developer to see the sql queries that are generated and sent for them as well as how many connections are established in order to serve those queries. A smaller number of connections is always preferable and in situations, such as the one shown in the example window below, where a large number of connections are established to service one transaction it is an indication that the action used to service this transaction may be a prime candidate for optimization. Some basic features of the SQL window are:

- A display of the number of connections associated with the request.
- The number of queries performed.
- The number of transactions that took place.
- The total time spent executing the queries.
- The total time the connections with the database remained open.
- The ordinal of the SQL query as well as the SQL query itself.
- The parameters used for the query.
- The records returned.
- The time spent executing the query.
- Whether the query was performed asynchronously.

Chrome 53.0

http://localhost:58287/Curriculum?lessonId=1289&returnLessonId=0

Cache

Configuration

Environment

Execution

Metadata

Model Binding

Request

Routes

SQL

Server

Session

Timeline

Tr

Ajax

History

SQL Statistics

# Connections	# Queries	# Transactions	Total query execution time	Total connection open time
51	51	1	3522.38 ms	3938.43 ms

Queries

Commands per Connection							Duration		
Ordinal	Command	Parameters				Records	Duration	Offset	Async
1	SELECT TOP (1) [Extent1].[UserId] AS [UserId], [Extent1].[InstitutionId] AS [InstitutionId], [Extent1].[UserName] AS [UserName], [Extent1].[FirstName] AS [FirstName], [Extent1].[LastName] AS [LastName], [Extent1].[DisplayName] AS [DisplayName], [Extent1].[EmailAddress] AS	Name	Value	Type	Size	1	57.67 ms	T+ 1786.26 ms	false
		@p_linq_0	betty	AnsiString	8000				
59.73 ms									

History Tab

This tab allows you to view the diagnostic data collected for previous requests made by your browser. Additionally it allows you to view requests made remotely by devices in disparate locations. According to the official Glimpse documentation, “This allows for several interesting use cases, including mobile debugging, remote client debugging, tracking workflows, etc.” A Case Study for collecting data from a remote browser can be located at <http://getglimpse.com/Docs/History-Tab>.

The layout of the tab is:

(List taken directly from Glimpse documentation)

- **Client Session Panel:** List of available client session's
- **Client Request Panel:** List of available client request's
- **Client Name:** Name given to the session that is being tracked by Glimpse
- **Method:** HTTP method used for the individual request (typically GET or POST)
- **Request URL:** URL that can be used identity individual requests
- **Browser:** Browser name that has been used for the request
- **Data/Time:** When the request was made
- **Ajax:** Whether the request was made via Ajax
- **Launch:** Opens a given request as the current context
- **Count:** Number of requests that are available to be viewed

Chrome 54.0 http://localhost:58287/DailyPlanning?selectedClassMeetingId=265&selectedClassSectionId=10&referenceClassSectionId=0										New update available		?	⌵	🔍																	
Cache		Configuration		Environment		Execution		Metadata		Model Binding		Request		Routes		SQL		Server		Session		Timeline		Trace		Views		Ajax		History	
Client		Count		View																											
Chrome 54.0		12		Inspect		Request URL				Method		Status Code		Duration		Date/Time				Is Ajax		View									
						http://localhost:58287/DailyPlanning?selectedClassMeetingId=265&select...				GET		200		2660.29 ms		10/29/2016 07:03:23				false		Inspect									
						http://localhost:58287/DailyPlanning/RemoveLesson?selectedLessonUsed...				GET		302		198.81 ms		10/29/2016 07:03:20				false		Inspect									
						http://localhost:58287/DailyPlanning?selectedClassMeetingId=265&select...				GET		200		2592.72 ms		10/29/2016 07:03:17				false		Inspect									
						http://localhost:58287/DailyPlanning/AddLesson?selectedLessonId=1288...				GET		302		313.88 ms		10/29/2016 07:03:14				false		Inspect									

Ajax Tab

Like the History tab, the Ajax tab allows you to view the recorded information for different requests. However, where the History Tab stores requests that cause the page to be (re)loaded, the Ajax tab stores the Ajax requests made by Javascript within your page that do not cause a page reload. In order to view the information about a particular Ajax request, click on the inspect link next to it. This will cause the context to switch to the selected Ajax request and cause all of the other tabs to display the data related to it. To reset the context back to the page content simply click on the 'Reset context back to the starting page' link. The columns in this tab are:

(List taken directly from Glimpse documentation)

- **Request URL:** The URL that was accessed by the Ajax request
- **Status:** Status of the resulting response from the server
- **Date/Time:** Date time of the request
- **Duration:** Approximate duration of the request
- **Is Async:** Whether the request was async/sync
- **Inspect:** Allows you to select the request you wish to inspect

NOTE: If you are using XML or JSON as your content type for an Ajax call, the request will show as crossed out. To remedy this, add the following to the Glimpse Configuration node.

```
<contentTypes>
  <add contentType="text/html" />
  <add contentType="text/xml" />
  <add contentType="application/json" />
</contentTypes>
```

Additional Resources

For more information about the tabs not discussed here as well as instructions on creating custom tabs refer to the official Glimpse Documentation. The official Glimpse Documentation provides a more thorough coverage of the customization and capabilities offered by Glimpse. and can be found at the following link. <http://getglimpse.com/Docs/>

Works Cited

"Documentation." *Glimpse*. Glimpse, 2016. Web. 9 Oct. 2016.
<<http://getglimpse.com/Docs/Execution-Tab>>.